

HW8

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1. (a) Let A be an upper-triangular matrix:

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ 0 & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & a_{nn} \end{bmatrix}$$

where for $i < j$, $a_{ij} = 0$. We will show that $a_{ii} = c_i \forall i \in \mathbb{Z}$, that the diagonal elements are the roots of the characteristic polynomial. The characteristic polynomial is defined as

$$p_A(X) := \det(X1_V - A) = \det \begin{bmatrix} X - a_{11} & -a_{12} & \dots & -a_{1n} \\ 0 & X - a_{22} & \dots & -a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & X - a_{nn} \end{bmatrix}$$

From the lecture notes, we know that the determinant of an upper-triangular matrix is the product of the elements along its diagonal. To prove this, recall the definition of the determinant as the sum over all permutations in the symmetric group S_n . For a term in the sum to be not equal to 0, we require $i \leq \sigma(i)$, $\forall i \leq n$. Note this implies that $n \leq \sigma(n)$, or $\sigma(n) = n$, since $\sigma(i)$ is at most n .

We proceed inductively, such that the term in the determinant that doesn't have an element under the diagonal implies that $\sigma(n-k) = n-k$ holds for all $k \geq 0$ up to n , we show that $\sigma(n-k-1) = n-k-1$. We already proved the base case $k = 0$. We know $n-(k+1) \leq \sigma(n-(k+1))$ since we don't want to use any of the elements below the diagonal as they are trivially zero, but by the inductive hypothesis, and since $\sigma(i)$ is injective (no two elements can map to the same element; otherwise it would not be a permutation), $\sigma(n-(k+1))$ cannot map to $n-k, n-(k-1), \dots, n$, since they have already been mapped to, so $\sigma(n-(k+1)) \leq n-(k+1)$. Combined with the fact we don't use any terms below the diagonal (since they would be trivially zero), $\sigma(n-(k+1)) = n-(k+1)$. So the non trivially zero term must be $(X - a_{11})(X - a_{22}) \dots (X - a_{nn})$, or the product of terms along the diagonal. Notice how this is already in the form of the splitting characteristic polynomial, in which the c_i terms are just the diagonal elements a_{ii} .

By Corrolary 6.5 in the lecture notes, the eigenvalues of the matrix A are the roots of the characteristic polynomial. By plugging in $X = a_{ii}$ into the characteristic polynomial, we see the $X - a_{11}$ term equal to 0, which causes the whole product to be equal 0, meaning a_{ii} is a root, which implies it is also an eigenvalue

- (b) Suppose A is a diagonalizable matrix under ordered basis $\mathcal{B}'$. Let D be the diagonal matrix of A with diagonal elements λ . Matrix D sends the ordered basis $\mathcal{B}$ of eigenvectors to the eigenvalues of λ . We use notation of $A = {}_{\mathcal{B}'}[A]_{\mathcal{B}'}$ and $D = {}_{\mathcal{B}}[D]_{\mathcal{B}}$. There exist change of basis matrices ${}_{\mathcal{B}'}[P]_{\mathcal{B}}$ such that

$$A = PDP^{-1}$$

But $P = {}_{\mathcal{B}'}[1]_{\mathcal{B}}$ and $D = \lambda \cdot {}_{\mathcal{B}}[1]_{\mathcal{B}}$. Putting this together

$${}_{\mathcal{B}'}[A]_{\mathcal{B}'} = {}_{\mathcal{B}'}[1]_{\mathcal{B}}(\lambda \cdot {}_{\mathcal{B}}[1]_{\mathcal{B}}){}_{\mathcal{B}}[1]_{\mathcal{B}'} = \lambda \cdot {}_{\mathcal{B}'}[1]_{\mathcal{B}'}$$

which proves that A is simply a scalar multiple of the identity matrix. In particular, it is a multiple of the eigenvalue itself.

- (c) Let's compute the characteristic polynomial of A first, suppose A is an $n \times n$ matrix. Since it is upper-triangular, the characteristic polynomial is simply the product of all $X - a_{ii}$. But since $a_{ii} = a$ for all i , we can get that the characteristic polynomial is

$$p_A(X) = (X - a)^n$$

This implies a is the only eigenvalue, which we determined from part (b) implies that the original matrix A is a scalar multiple of the identity matrix if we assume it is diagonalizable, implying that off diagonal elements must be zero. So if there is a nonzero off diagonal element, there is a contradiction, and it can't be diagonalizable

2. (a) We evaluate the characteristic polynomial.

$$p(X) = \det \begin{pmatrix} X-5 & -19 & 0 \\ 1 & X+3 & 0 \\ 5 & -2 & X-5 \end{pmatrix} = (X-5)[(X+3)(X-5) + 19]$$

This is obtained from Laplace Cofactor Expansion on the third column. We look for roots to this characteristic polynomial, expanding:

$$p(X) = (X-5)[(X^2 - 2X + 4)]$$

There is only a single root in $\mathbb{R}$ for this polynomial, $X = 5$. We now find three linearly independent eigenvectors (if possible). Suppose (u, v, z) is an eigenvector.

$$\begin{pmatrix} 0 & -19 & 0 \\ 1 & 8 & 0 \\ 5 & -2 & 0 \end{pmatrix} \begin{pmatrix} u \\ v \\ z \end{pmatrix} = \begin{pmatrix} -19v \\ u + 8v \\ 5u - 2v \end{pmatrix} = u \begin{pmatrix} 0 \\ 1 \\ 5 \end{pmatrix} + v \begin{pmatrix} -19 \\ 8 \\ -2 \end{pmatrix}$$

which implies the $\text{rank}(5 \cdot 1_V - A) = 2$. By rank-nullity, we figure out the dimension of the 5-eigenspace, which is simply the kernel, is equal to $3 - 2 = 1$. We proved in the notes that the sum of geometric multiplicities equals the dimension of the vector space, 3. But $1 \neq 3$, so the matrix is not diagonalizable. Alternatively, note that for $(5 \cdot 1_V - A)v_1 = 0$, we require $v = 0$, $u = 0$, leaving the only free choice to be z , which implies the dimension of the 5-eigenspace is 1. Alternatively, note the characteristic polynomial doesn't split over $\mathbb{R}$, so it cannot be diagonalizable anyways.

- (b) We evaluate the characteristic polynomial

$$p(X) = \det \begin{pmatrix} X-2 & -7 & -1 \\ 0 & X-2 & 3 \\ 0 & 0 & X-5 \end{pmatrix} = (X-2)^2(X-5)$$

The roots are $X = 2, 5$ with algebraic multiplicity 2 and 1 respectively. We look for eigenvectors for the eigenvalue 2, suppose the eigenvector to be of form (u_1, v_1, w_1)

$$\begin{pmatrix} 0 & -7 & -1 \\ 0 & 0 & 3 \\ 0 & 0 & -3 \end{pmatrix} \begin{pmatrix} u_1 \\ v_1 \\ w_1 \end{pmatrix} = \begin{pmatrix} -7v_1 - w_1 \\ 3w_1 \\ -3w_1 \end{pmatrix}$$

Then $w_1 = 0$, $v_1 = 0$, so the only free choice we have is u_1 . So the 2-eigenspace is spanned by $(1, 0, 0)$, which has dimension 1, which is not equal to the algebraic multiplicity of 2, so the matrix is not diagonalizable either.

3. (a) We first understand how T acts on the standard ordered basis defined as $\mathcal{B}' = (1, x, x^2)$. Evaluating at each basis element,

$$T(1) = 0,$$

$$T(x) = 1,$$

$$T(x^2) = 2x + 4.$$

So the matrix representation in the ordered basis $\mathcal{B}'$ is

$${}_{\mathcal{B}'}[T]_{\mathcal{B}'} = \begin{pmatrix} 0 & 1 & 4 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{pmatrix}.$$

The characteristic polynomial for this operator is

$$p_T(X) = \det \begin{pmatrix} X & -1 & -4 \\ 0 & X & -2 \\ 0 & 0 & X \end{pmatrix} = X^3$$

The eigenvalue is $X = 0$ with algebraic multiplicity 3. Let's find the 0-eigenspace, let (u_1, v_1, w_1) be the form of an eigenvector.

$$\begin{pmatrix} 0 & -1 & -4 \\ 0 & 0 & -2 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} u_1 \\ v_1 \\ w_1 \end{pmatrix} = \begin{pmatrix} -v_1 - 4w_1 \\ -2w_1 \\ 0 \end{pmatrix} = 0$$

so $v_1 = 0$, $w_1 = 0$, meaning the eigenvector is $(u_1, 0, 0)$ which is spanned by $(1, 0, 0)$, so it has geometric multiplicity 1 which is not equal to 3, so this operator is not diagonalizable.

- (b) Let us see how T takes the ordered basis $\mathcal{B}' = ((1, 0), (0, 1))$

$$T(1, 0) = (1, -i)$$

$$T(0, 1) = (-i, 1)$$

so the matrix representation in the ordered basis $\mathcal{B}'$ is

$${}_{\mathcal{B}'}[T]_{\mathcal{B}'} = \begin{pmatrix} 1 & -i \\ -i & 1 \end{pmatrix}.$$

The characteristic polynomial is then

$$p_T(X) = \det \begin{pmatrix} X - 1 & i \\ i & X - 1 \end{pmatrix} = (X - 1)^2 + 1 = X^2 - 2X + 2$$

The roots/eigenvalues of the characteristic polynomial are

$$X = 1 \pm i$$

We search for a eigenvector of $1 + i$ first, letting this eigenvector be of the form (u_1, v_1) , $u_1, v_1 \in \mathbb{C}$

$$\begin{pmatrix} i & i \\ i & i \end{pmatrix} \begin{pmatrix} u_1 \\ v_1 \end{pmatrix} = \begin{pmatrix} iu_1 + iv_1 \\ iu_1 + iv_1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

which implies that $u_1 = -v_1$, or that the eigenvector is of form $(-v_1, v_1) = v_1(-1, 1)$ which means $\dim \ker((1 + i)1_V - A) = 1$ spanned by a single vector, meaning the the geometric multiplicity is equal to the algebraic multiplicity.

For the conjugate eigenvalue $X = 1 - i$, the process is identical. Letting (u_2, v_2) be an eigenvector,

$$\begin{pmatrix} -i & i \\ i & -i \end{pmatrix} \begin{pmatrix} u_2 \\ v_2 \end{pmatrix} = \begin{pmatrix} -iu_2 + iv_2 \\ iu_2 - iv_2 \end{pmatrix}$$

which implies $u_2 = v_2$, or the eigenspace is $v_1(1, 1)$ which is spanned by one vector, so it has dimension 1, which equals the algebraic multiplicity. So since the algebraic multiplicity equals the geometric multiplicity for all eigenvalues, this operator is diagonalizable. The corresponding diagonal matrix is simply the diagonal matrix where the diagonal elements are the eigenvalues.

$$D = \begin{pmatrix} 1+i & 0 \\ 0 & 1-i \end{pmatrix}$$

If we define a new ordered basis to be eigenvectors, $\mathcal{B} = ((1, -1), (1, 1))$, we can obtain a change of basis matrix P from $\mathcal{B}$ to $\mathcal{B}'$ to be

$$P = \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$$

such that

$${}_{\mathcal{B}'}[T]_{\mathcal{B}'} = PDP^{-1}$$

4. Following the hint,

$$\begin{aligned} (T - \lambda_1)(v_1 + v_2 + \dots + v_m) &= (T - \lambda_1)v_1 + \dots + (T - \lambda_1)v_m \\ &= Tv_1 + \dots + Tv_m - \lambda_1v_1 - \dots - \lambda_1v_m = (\lambda_2 - \lambda_1)v_2 + \dots + (\lambda_m - \lambda_1)v_m \end{aligned}$$

Note this shrinks the size of a sum of m vectors to a sum of $m - 1$ vectors, motivating induction. Let's start with a single vector, and build up to m . The base case when $m = 1$ is true since if $v_1 \in W$, then $v_1 \in W$. Now let's assume that any sum of $m - 1$ eigenvectors being in W implies that those $m - 1$ vectors lie in W . Suppose we have a sum of m eigenvectors $v_1 + \dots + v_m \in W$. If we apply $T - \lambda_1$ on this sum, we get something back in W because

$$(T - \lambda_1)(v_1 + \dots + v_m) = T(v_1 + \dots + v_m) - \lambda_1(v_1 + \dots + v_m)$$

in which the first term is in W since T is W -invariant, and the second term is closed by scalar multiplication, so applying $T - \lambda$ on the sum keeps the result in W .

As we determined earlier, applying $T - \lambda_1$ on the sum of m vectors outputs a sum of $m - 1$ vectors. However, since a scalar multiple of an eigenvector is still an eigenvector, so we can define a new set of eigenvectors $u_i = (\lambda_i - \lambda_1)v_i$. This means

$$(T - \lambda_1)(v_1 + v_2 + \dots + v_m) = u_2 + \dots + u_m \in W$$

By inductive hypothesis, this means $u_i \in W$, which means. Since eigenvalues are distinct, $(\lambda_i - \lambda_1)^{-1}$ exists, so we can solve for $v_i = (\lambda_i - \lambda_1)^{-1}u_i$ and by closure of scalar multiplication, implies that $v_i \in W$ for $i \neq 1$. Returning the original sum, we now know $v_2 + \dots + v_m \in W$, and that $v_1 + v_2 + \dots + v_m \in W$. This must imply that $v_1 \in W$, since we can add the additive inverse of $v_2 + \dots + v_m$ to show $v_1 \in W$.

5. Since T is diagonalizable, we can write any vector $v \in V$ as a sum of vectors

$$v = v_1 + \dots + v_k$$

where v_i is a vector in the λ_i -eigenspace, made possible since the eigenvectors form a basis. Since W is a subspace of V , we can say the same about any vector in W , that is w can be written as a sum of eigenvectors from the λ_i -eigenspace.

$$w = v_1 + \dots + v_k$$

By the result in problem 4, $w \in W$ and T is W -invariant implies that $v_1, \dots, v_k \in W$. So each v_i is actually an eigenvector for the restricted operator $T|_W$. Note, this actually also implies that W is spanned by the eigenspaces of $T|_W$. We now show the the union of all eigenvectors are linearly independent.

Let $\lambda_1, \dots, \lambda_r$ be eigenvalues of the restricted operator $T|_W$, and $\mathcal{B}_1, \dots, \mathcal{B}_r$ be the basis of the respective eigenspaces. The geometric multiplicity of eigenvalue λ_i is simply $\mathfrak{g}_{\lambda_i} = |\mathcal{B}_i|$ by definition. By Lemma 2.7 in the lecture notes, we get that the set

$$\bigcup \mathcal{B}_i$$

is linearly independent in W . But these are also all the eigenspaces of W , which we've shown span W . Putting this all together, the union of all $\mathcal{B}_i$ is a basis of W , or

$$\sum |\mathcal{B}_i| = \sum \mathfrak{g}_{\lambda_i} = \dim W$$

which by Theorem 2.9, implies that $T|_W$ is diagonalizable.

6. We proceed by induction on n , where the matrix A is an $n \times n$ matrix. The case $n = 1$ is trivially true, a single element matrix is trivially an upper-triangular matrix.

Now assume any $(n - 1) \times (n - 1)$ matrix whose characteristic polynomial splits is conjugate to an upper-triangular matrix. Let A be an $n \times n$ matrix. The characteristic polynomial for A splits, so there exists at least one root/eigenvalue, λ_1 . Let v_1 be the eigenvector corresponding to that root. We can extend v_1 to an ordered basis $\mathcal{B} = (v_1, \dots, v_n)$, such that when we transform the matrix A into this basis, we obtain $(\lambda, 0, \dots, 0)$ as the first column. Formally, suppose A is originally expressed in terms of a basis $\mathcal{B}'$, then

$$\mathcal{B}[A]\mathcal{B} = \mathcal{B}[1]_{\mathcal{B}'\mathcal{B}'}[A]_{\mathcal{B}'\mathcal{B}'}[1]_{\mathcal{B}} = \begin{pmatrix} \lambda_1 & * \\ 0 & A' \end{pmatrix}$$

where A' is a $(n - 1) \times (n - 1)$ matrix. Now the characteristic polynomial for A is

$$p_A(X) = (X - \lambda)p_{A'}(X)$$

but since $p_A(X)$ splits, then so must $p_{A'}(X)$. Recall, the characteristic polynomial does not depend on choice for basis either, so using A' is valid. Now, by the inductive hypothesis, A' is conjugate to an upper-triangular matrix,

$$A'' = PA'P^{-1}$$

But we need to somehow rewrite A such that we can substitute A'' into the bottom right corner element. This can be done by another conjugation

$$\begin{pmatrix} 1 & 0 \\ 0 & P \end{pmatrix} \mathcal{B}[A]\mathcal{B} \begin{pmatrix} 1 & 0 \\ 0 & P^{-1} \end{pmatrix} = \begin{pmatrix} \lambda_1 & * \\ 0 & PA'P^{-1} \end{pmatrix} = \begin{pmatrix} \lambda_1 & * \\ 0 & A'' \end{pmatrix}$$

which is now an upper-triangular matrix, proving the statement, since conjugates are transitive.